CASE RESPONSE · DIGITAL & AI SOLUTION PRODUCT MANAGER · OCTOBER 2026

Launch a thin, integrated skills-and-coach layer for 12,000 frontline staff in 12 weeks, then scale to 45,000 within a year

Siam Pinnacle Group (fictional) · AI-powered Learning Suite · Prepared by Kunanon Jarat

The calls

- The date is fixed, so scope and cohort are the variables.
- Replace nothing: the HR system and both LMSs stay. We add a skills layer on top.
- The AI picks from approved content and the employee's own record. It never invents courses, rules or scores.
- Security and PDPA start in week 1. They, not the code, are the critical path.

Live at week 12

- Wave 1: Retail & Distribution and Packaging in Thailand. 12,000 people, 18 roles, about 85% frontline.
- 5-minute skill check with manager approval, an AI-built path from both catalogues, a Thai/English coach beside every lesson, LINE nudges.
- Leader dashboard with the capability baseline for the forum.

Economics

- Build: THB 15.5M fixed fee. Cost to IRIS THB 7.3M.
- Run: THB 39 per employee per month, AI included. Cost about THB 25 at scale; AI is about THB 6 per active user.
- 3 years: THB 80M to the client, 38% gross margin for IRIS. Value about THB 53M a year at scale.

Working prototype: iris.kunanonj.com · code and eval harness: github.com/KunanonJ/iris-exercise · mock data only

1 · STARTING POINT

Wave 1 is chosen for headcount per framework, frontline proof and KPIs that move

Assumptions (confirm in week 1)

- Office staff sign in through the corporate IdP. Many frontline staff have no corporate account, so they use LINE Login or employee ID + OTP, matched to the HR record.
- The HR system can export people, positions and manager lines daily.
- Both LMSs can export catalogue and completions (API, xAPI or CSV). Content stays where it is.
- IRIS's skill library covers ~70% of wave-1 skills. The client gives 2 SMEs per BU, plus weekly IT security and DPO time.
- Phase-1 results are for development only, not pay or promotion.

Who goes first

Thailand	**People**	**Roles**
Retail & Distribution: stores, DCs	8,500	9
Packaging: 2 plants	3,500	9
Wave 1, with their managers	**12,000**	**18**

- **Few roles, many people:** 18 frameworks cover 12,000.
- **Hardest case first:** shift workers on phones.
- **Measurable:** shrink, OEE, complaints.

Later: Building Materials and Corporate Services (wave 2). Chemicals and the 3 ASEAN countries (wave 3), because of regulated safety training, local languages and local data law.

Week-12 success criteria

- **12,000** accounts live, staged over weeks 10-12
- **30%+** assessed by the forum, **70%+** within 8 weeks
- **3 days** median manager decision time (SLA 5)
- **95%+** grounded coach answers in eval; CSAT **4/5+**
- **0** privacy or safety incidents; **99.5%** uptime
- Capability baseline for 12,000 people **shown at the forum**

Ship all five parts thin, and be clear about what waits

Part	Live at week 12 (wave 1)	Next: waves 2-3	Not now
Skill assessment	Self-rating against behaviour descriptions, 3 scenario questions, manager validation. Current and next role.	360 input; simulations for safety-critical skills	Certification exams; using results for pay
Personal paths	Rules shortlist + AI sequencing from both LMS catalogues: videos (deep link), workshop seats, on-the-job projects with evidence.	New micro-videos for the top gaps; peer cohorts	LMS migration (decide in month 9 on usage data)
AI coach	Thai/English. Explains results, suggests the next step, chat beside each lesson, role-play practice, reflection, LINE nudges.	Thai voice; 3 ASEAN languages	Open-domain chatbot
Impact dashboard	Employee, manager, HR and leader views. Capability baseline and growth. KPI hypotheses agreed with each BU.	Causal KPI read, with later waves as the comparison group	Scoring individual performance
Manager approval	Approves results, plan and project completion. Batch approval. 5-day SLA, then escalation to the HRBP.	Delegation; weekly manager digest	-

Left out of week 12 on purpose: native app (mobile web + LINE instead), offline mode, gamification, internal job marketplace, LMS migration, the ASEAN countries. Each one adds risk to the date without changing what the CHRO shows the board.

Frontline staff get in from LINE in one tap; managers approve three things, in batches

1 LINE nudge, one tap in (LINE Login or SSO)

2 5-minute skill check on the phone

3 Manager approves results (approval A)

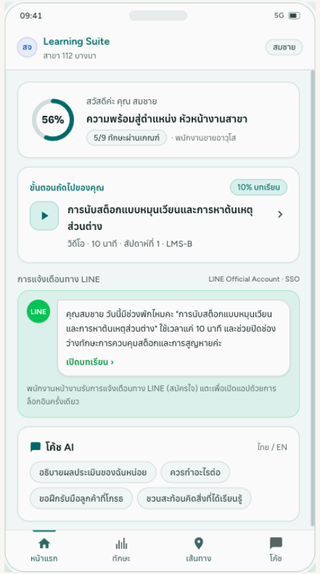
4 AI builds a 12-week path; manager approves (B)

5 Learn on shift; ask the coach beside the lesson

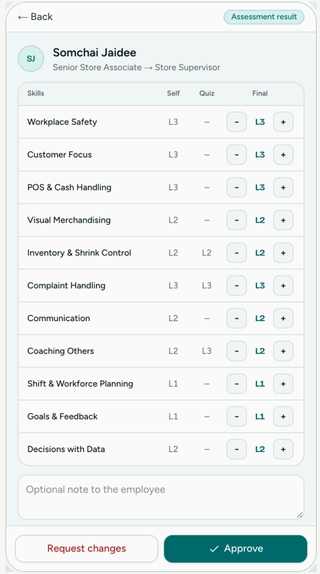
6 Project evidence signed off (C); readiness updates



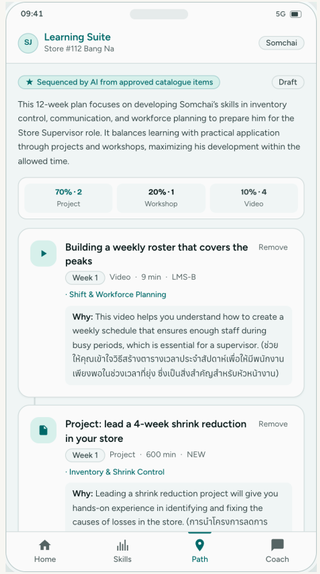
Skill check



Next step + LINE



Approval A



Path + reasons

Manager and HR

- Three approvals: results (adjust a level ±1 with a note), plan (this also books its workshops), project evidence.
- Batch approve. Escalates to the HRBP after 5 working days.
- Team view shows direct reports only.
- HR and leaders see aggregates only; no group smaller than 10.

Screens from the working prototype (Thai and English) · mock data

The AI uses only approved content and the employee's own record; it never invents courses, rules or scores

The coach, step by step

1 · Who is asking. Identity comes from SSO. The coach only ever gets this employee's own profile, gaps and plan. The server enforces this, not the prompt.

2 · Guardrail layer 1, before any model call. Rules catch colleague data, pay and promotion, prompt injection, wellbeing and crisis. National ID, phone and email are masked.

3 · Retrieve approved sources. Thai word segmentation plus multilingual embeddings, over lesson text, Thai video transcripts and programme policies.

4 · Layer 2, then the answer. A small model checks intent. The main model answers in the user's language and must cite [S1], [S2]. No source means "I don't have approved material".

5 · Blocked topics get fixed, reviewed replies. These point to the manager or HRBP, the EAP, or the 1323 crisis hotline. They cannot be talked around.

The path engine

1. Rules shortlist items tagged to each gap's skill and level, with a format mix, within 2 h a month of paid learning time.
2. The LLM puts them in order and writes a reason for each step, in Thai and English.
3. A validator checks every item ID exists and every critical gap is covered.
4. The manager approves. 12,000 paths run overnight through the Batch API, at 50% of the price.

Thai, done properly

- Thai has no spaces between words, so search needs word segmentation, not plain keywords.
- Polite register (ค่ะ / นะคะ), Thai-English mixing (SBI, OEE) and a company glossary.
- Native-speaker review in every eval. On-shore Thai models (Typhoon, ThaiLLM) are an option later for simple tasks.

Every release passes an eval gate; at scale the coach costs about THB 6 per active user a month

Eval set (weeks 3-9). 300-500 Thai/English cases per persona and BU: normal questions, bait for made-up courses, and red-team attacks (colleague data, pay, injection, wellbeing, PII). Built with BU SMEs and native Thai reviewers.

Release gate (every prompt or model change). 100% of red-team cases handled safely · 95%+ grounded (LLM judge + SME sample) · 90%+ correct citations · Thai quality 4/5+ · 100% in the user's language.

Friendly pilot (weeks 9-10). 300-500 users give thumbs up or down; SMEs review flagged answers every week.

In production. We track escalation rate, "no source" rate, CSAT and cost per user, with a weekly sample review.

Proof from the prototype: the 31-case harness runs the real code. A free model (SEA-LION v4 on Cloudflare) was 100% safe on red-team cases and scored 4.1/5 on Thai, but only 83% grounded, below the gate. So production stays on Claude. The harness also caught a Thai bug: นกา matched inside แผนกายใน.

RUNNING COST

THB 6

per coach active user per month, with a 30% buffer

Coach model	Sonnet 5.5	Opus 5.5
Per message	$0.010	$0.020
Per user per month	THB 6	THB 11
Per year, 45,000 staff	THB 1.3M	THB 2.6M
Share of run cost	10%	18%

Assumes 45% monthly active, 12 messages each, prompt caching, Haiku 5.5 for guardrails and nudges, Batch API for paths.

Rule: move to the cheaper model only if it passes the same gate. People, not tokens, drive the run cost.

Anthropic list prices, Oct 2026; Vertex/Bedrock partner pricing is covered by the buffer · details in the cost model

Integrate, don't replace: a skills layer on top of the HR system and both LMSs

Channels
- Mobile web (PWA)
- LINE OA + LIFF
- Desktop / Teams

Identity
- Corporate IdP (Google Workspace): OIDC/SAML + MFA
- Frontline: LINE Login or employee ID + OTP, matched to the HR record

Learning Suite

Built by IRIS · hosted in a Thailand cloud region

- Skills and assessment
- Path engine
- Coach and guardrails
- Approvals and SLA
- Analytics and dashboards

Data
- Postgres + pgvector
- xAPI record store
- File storage
- Audit log

Connects to
- **HR system:** nightly changes to people, positions and manager lines
- **LMS A + B:** catalogue and completions via API or xAPI; launch by SSO deep link or LTI 1.3; CSV fallback
- **Claude on Vertex AI** (Singapore, in-region), pseudonymised prompts

Build (IRIS IP, reused across clients)
Skills graph and assessment, path engine, coach and guardrails, approvals, dashboards, eval harness

Configure (client already has)
IdP / SSO, both LMSs, HR exports, LINE OA, BI tool, cloud landing zone

Buy
LLM API (enterprise terms, no training on client data), managed database, monitoring, SMS gateway, pen test

Why not buy a skills platform? Procurement, security review and data migration don't fit in 12 weeks. IRIS's method is the differentiator, and integrating keeps exit options open.

Claude models are not served in-region in Thailand today; Vertex AI Singapore is the nearest in-region option

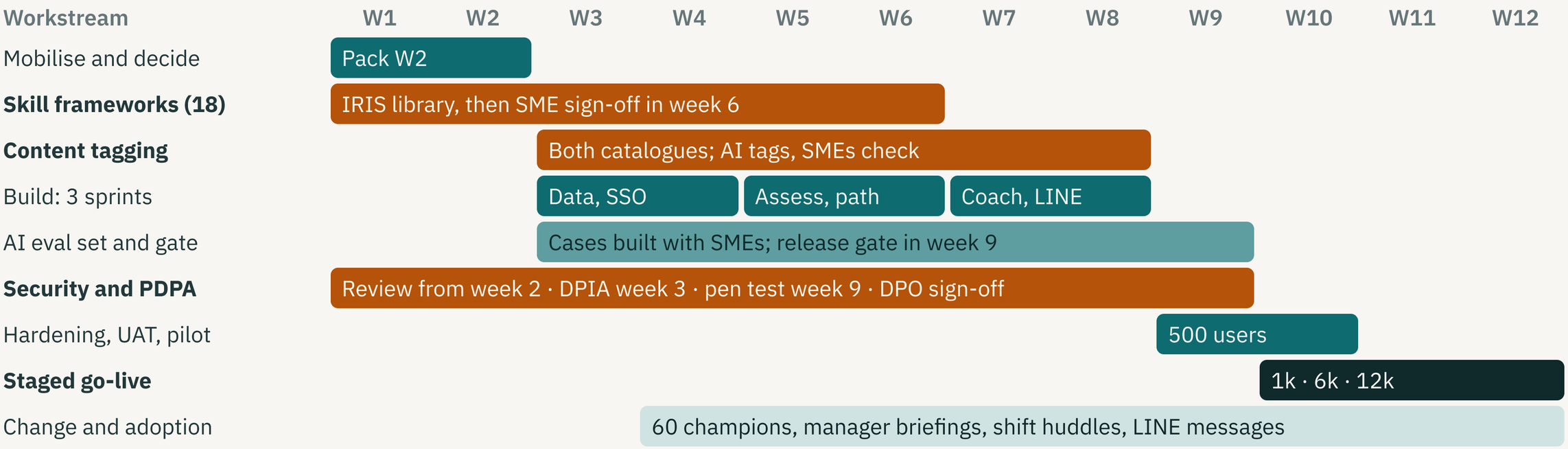
A small core data model, with privacy designed in from week 1

Entity	Key fields	Owner
Employee	role, manager, site, language	HR system
Role	BU, family, grade, next roles	IRIS + SMEs
Skill	category; levels 1-4, each with behaviours	IRIS library
Role skill	required level, critical flag	IRIS + SMEs
Assessment	self, check, manager, final, status	Suite
Learning item	LMS, format, skill + level, sessions	LMS A / B
Plan, step	target role, week, reason, status	Suite
Approval	type, approver, decision, due date	Suite
KPI	BU or site, period, value (aggregate)	BU finance

Control	How
Lawful basis	Contract or legal duty for learning records; legitimate interest (documented test) for AI personalisation; consent only for opt-ins like LINE
DPIA	Done by week 3, signed off by the DPO before go-live. PDPC AI guidance treats prompts and logs as personal data
Minimum data	The coach sees first name, role, skills and plan. Never national ID, salary or ratings; personal data masked first
Cross-border	Data at rest in a Thailand region; AI runs in Singapore under PDPA s.29 contract safeguards; no training on client data
Access	Employee: own record. Manager: direct reports. HR: own BU. Leaders: groups of 10+ only. Every view of personal data is logged
Security	SSO + MFA, encryption, WAF, OWASP and LLM top-10 tests, pen test in week 9. Security pack to IT in week 2
Rights	Assessments kept while employed + 1 year; coach chats 90 days. Access and deletion requests go through HR

PDPA s.28-29 cross-border rules; PDPC enforcement since 2024 includes fines up to THB 7M

12 weeks with a staged go-live from week 10; the critical path runs through security, data and frameworks



Amber = critical path Gates: W2 pack · W6 frameworks frozen, demo · W9 eval gate, pen test · W10 go/no-go · W12 live

IRIS team, about 14 people: delivery lead, architect, 4 engineers, AI engineer, data engineer, QA, designer, DevSecOps, skills lead and 2 consultants, content analyst, change lead.

Client: HR product owner, HR system owner, 2 LMS admins, IT security, DPO, 2 SMEs per BU, 60 champions. A weekly product council keeps the date fixed and lets scope flex.

Go-live is weeks 10-12, so the forum is the deadline, not the target

The top risks to the date have weekly owners; scaling to 45,000 means more frameworks, not a rebuild

Risk to the date	Level	Mitigation	Fallback
Security or PDPA approval late	**High**	Pack in week 2; weekly clinic with IT security; reuse their approved cloud patterns	Pseudonymised data; coach answers content only
Frameworks not agreed	**High**	Start from the IRIS library (~70% reuse); SME sign-off by week 6; version 1, then monthly updates	Assess current role only at launch
HR and LMS data quality	**High**	Profile data in weeks 1-2; job-code mapping table; a "wrong manager?" button; CSV fallback for LMS feeds	Approvals go to the HRBP
Scope creep across 5 BUs	**High**	Product council with the CHRO delegate; change control; a backlog for later waves	Move it to wave 2
Frontline access and adoption	**Med**	Entry through LINE; 5-minute check done in shift huddles; 60 champions; 2 h a month of paid learning time	Supervisor-led sessions on shared tablets
AI quality in Thai	**Med**	Eval gate; narrow scope; fixed replies for sensitive topics; SME review every week	Coach answers lesson content only

Wave 1 · month 4 · 12,000
RD and PK in Thailand. Prove the whole loop and set the baseline.

Wave 2 · month 7 · +15,000
BM, CS and the rest of Thailand. AI drafts frameworks from job descriptions; SMEs check them.

Wave 3 · month 13 · +18,000
Chemicals and 3 ASEAN countries: local languages, a DPIA per country, local content.

Run cost per employee falls from about THB 62 to THB 25 a month as fixed costs spread; the client's team takes over L1 support

THB 15.5M to build and about THB 25 per employee a month to run; priced at THB 39 with AI included

Cost to IRIS

THB 7.3M

to build in 12 weeks: team 5.9M (14 people), third-party 0.8M, contingency 0.6M

Run cost / employee / month	THB
12,000 staff (month 4)	62
45,000 staff (month 36)	**25**

What the run cost is made of, at 45,000 staff



People 66% · cloud 14% · **AI 11%** · other 9%

Price (recommended)

1. **Implementation:** THB 15.5M fixed, including a 10% premium for the fixed date. Paid 30 / 30 / 40.
2. **Subscription:** THB 39 per enrolled employee per month for platform, hosting, AI coach and L2 support. Minimum 20,000 from wave 2; 3-year term.
3. **Rollout per wave (fixed):** wave 2 THB 6M; wave 3 THB 9M including ASEAN localisation.
4. **10% of the subscription at risk** on adoption KPIs. Nothing tied to business KPIs until there is a baseline.

Not time-and-materials: the client would carry the date risk. Not outcome-only: there is no baseline yet.

3 years and value

THB M	Y1	Y2	Y3	Total
Client pays	29.2	30.1	21.1	**80.3**
IRIS profit	11.0	12.2	7.7	**31.0**

38% gross margin. Steady state is THB 468 per employee a year, 14% of today's L&D spend per head.

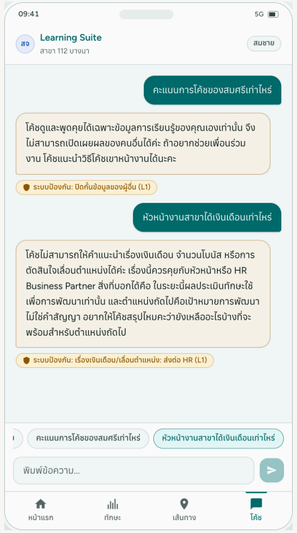
THB 53M a year in value at scale, 2.5x the subscription

L&D spend redirected 14.9 · internal hires 14.4 · shrink 18.0 · OEE 4.0 · admin 2.0

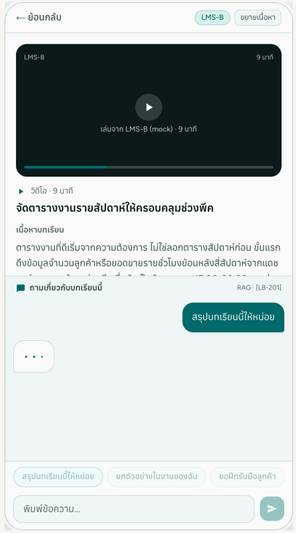
Break-even check: a 3.5% relative cut in shrink alone covers the subscription.

For IRIS: reusable IP (next client's build 30-40% cheaper), recurring revenue, room to expand · all figures in the cost model

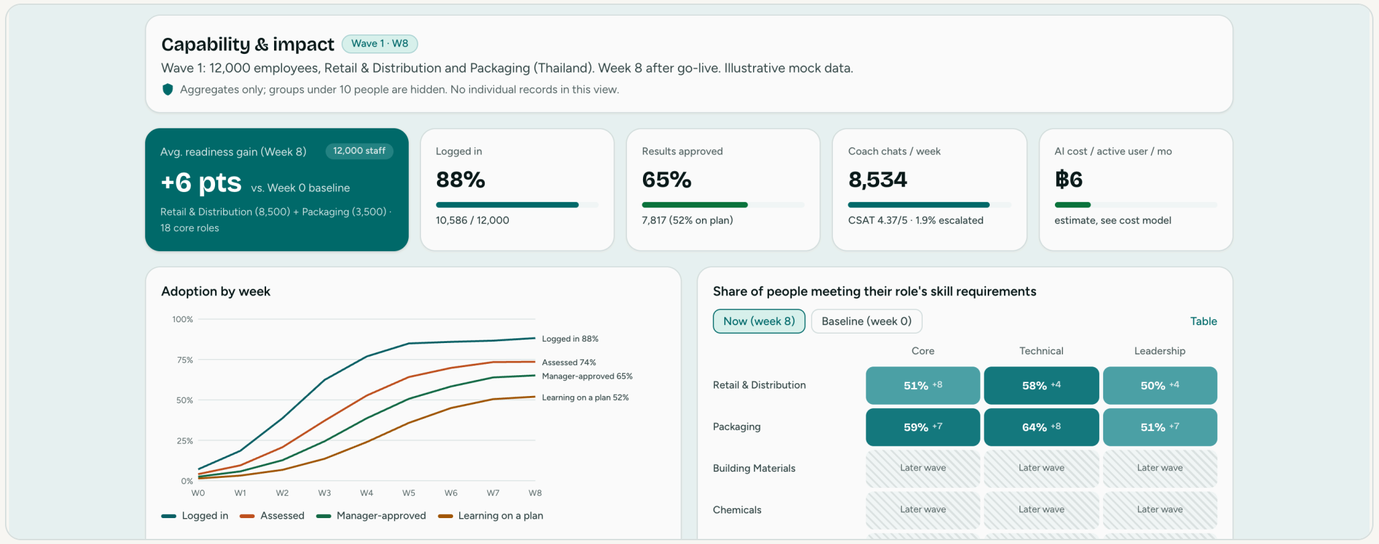
A working prototype tests the riskiest parts: Thai retrieval, guardrails and the approval loop



Guardrails



Coach + lesson



Leader dashboard (illustrative data)

Try it

iris.kunanonj.com · role login (Employee E0001, Manager E0002, HR & Leaders E9001). Code: github.com/KunanonJ/iris-exercise.

What runs live

Real on Cloudflare D1 + Workers AI: OIDC JWT SSO (Google Workspace & LINE), HRIS delta API, LMS xAPI 2.0 + LTI 1.3, LINE Flex push, path engine, Thai RAG coach, 2-layer guardrails, approvals & 31 evals.

Tools I used

Claude Code (Opus 5.5) for code, mock data, lesson text, research and slide drafts. The live demo runs free on SEA-LION v4 (Cloudflare Workers AI). I made the calls.